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Explainable artificial intelligence to estimate the Sri Lankan (Ceylon) Tea crop yield

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ABSTRACT

Accurate cash crop yield prediction is important in the context of changing climate to ensure sustainable development. Artificial intelligence (AI) has played a significant role in prediction in nonlinear systems including cash crop prediction. However, despite its global recognition, Sri Lanka's renowned Ceylon Tea has never been modeled for yield prediction based on meteorological, soil, and fertilizer parameters. Sri Lanka is one of the most impacted countries due to climate change while its agricultural production is significant in economic contribution. Therefore, this study presents the first-ever research to predict the tea crop yield with respect to meteorological, soil, and fertilizer features. State-of-the-art machine learning models were tested with the aid of Dampahala Tea Company production. Results showcased that the CatBoost algorithm produces the best prediction model with a coefficient of determination, $R^2_{CatBoost}=$ of 0.90374 (other models tested, $R^2_{XGBoost}=$ 0.87385; $R^2_{LightGBM}=0.79772$; $R^2_{ANN}=0.57346$; $R^2_{Ridge}=0.42033$; $R^2_{LASSO}=0.42512$; $R^2_{ElasticNet}=0.42168$). In addition, explainability aspects using SHAP showcased that the morning relative humidity, evaporation, and T0 200 fertilizer have a significant impact on the prediction model. The developed AI model can be disseminated across the country to enhance understanding of predictions under future climatic scenarios.

1. Introduction

Climate action, the 13th of the Sustainable Development Goals has given significant attention to the food production and food sustainability of the world. Agricultural production in many countries has been adversely impacted not only due to the drought conditions [1–3] but also the intensified precipitation conditions [4,5]. Therefore, while investigating countermeasures for climate change, further research is being carried out to develop climate-resilient varieties of crops that can be grown for future needs [6,7]. However, replacing climate resilience crops for mega-scale plantations such as tea, rubber, coconut, and other cash crops (which are grown primarily for commercial sale rather than local consumption) is difficult at least for developing countries. Therefore, it is crucial to understand the impact of climate change on agricultural production within these existing large-scale plantations.

Climate change not only impacted agricultural food production but also there is enough evidence to see the impact on the quality of the products [8,9]. Literature showcases evidence of changes in taste [10] and nutrition levels [11]. Therefore, food security is a significant threat. In addition, the economic situation of most developing countries is based on cash crops [12–14]. Therefore, on top of the country's food security, the economic stability has shattered due to the impact of climate change on agriculture.

Tea, whether it is black tea or green tea, is one of the most widely consumed beverages in the world [15]. Among tea-producing countries, China, India, Kenya, and Sri Lanka are major producers and exporters [16]. Tea unlike many other beverages, the tea's taste and quality have a unique relationship to the climatic parameters [17,18]. Atmospheric

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temperature, rainfall, relative humidity, altitude, soil condition, and wind are some of the parameters that influence the growth of the tea plant and the quality of the production. Due to its unique geographical features and best climatic factors, Ceylon Tea, in other words, Sri Lankan tea holds the world's best quality since it is inhabited in Sri Lanka. Therefore, Sri Lankan tea is in high demand in the international market.

Consequently, Ceylon tea is in high demand in the international market. In 2015, Sri Lanka earned over 425 billion Sri Lankan rupees (approximately 1.3 billion USD) from tea production, making a significant contribution to the national economy [19]. Middle Eastern countries and the Russian Federation are the two biggest buyers of Sri Lankan tea [20]. Therefore, it is highly important to maintain the quality while keeping production up to the required demand. However, Sri Lanka is rated as one of the most highly impacted islands by the ongoing climate change [21]. Tea crop production has been significantly impacted in some of the areas during extreme weather conditions [22,23]. Therefore, it would be highly important to understand the relationship between tea crop production and ongoing climate change. Despite this importance, a comprehensive study covering all of Sri Lanka has yet to be conducted.

The literature presents many studies in crop yield prediction with respect to climate change [24–30]. Among various simulation models, machine learning techniques are frequently used to understand the nonlinearity and then to predict the tea crop yield.

Batool et al., [24] presented a hybrid technique in combining crop simulation models with machine learning techniques to predict tea yields. Their case study focused on the Pakhal Valley Shinkiari, Mansehra, Pakistan. Machine learning models were developed that integrate environmental factors such as temperature, precipitation, and soil conditions in the prediction of tea crop yield. Their findings demonstrated the potential for more reliable yield forecasting, benefiting tea growers in adapting to changing environmental conditions.

Several similar studies can be found in the literature along the lines of predicting tea crop yield using machine learning concepts [31,32]. In addition, some researchers have employed satellite-based image processing techniques in collaboration with machine learning techniques to predict the tea crop yield [33–36]. Furthermore, machine learning approaches have been continuously applied in tea related research, such as location identification [37], assessment of taste and quality of tea [38, 39], etc.

However, even in the era of artificial intelligence, the world's best quality tea producer [40], Sri Lanka, is limited with related research in predicting tea yield under climate change. The Tea Board of Sri Lanka reports that Sri Lanka exports its tea to more than 140 countries worldwide (more information is at https://srilankateaboard.lk/). There were several attempts in literature; however, none of them looked at state-of-the-art machine learning approaches for accurate predictions. Jayatilake and Rankothge [41] utilized regression-based models to examine tea yield in relation to weather data specifically for the Uva province, Sri Lanka. However, they were unable to find a comprehensive analysis of tea yield predictions in the context of changing climate conditions. Nevertheless, a few related research works have showcased the impact of changing climate not only in yield change but also in quality changes [16].

Therefore, there is a greater potential to explore more artificial intelligence to understand its applications in tea crop yield predictions in Sri Lanka. Thus, this paper presents an explainable artificial intelligence in predicting tea crop yield using meteorological, soil, and fertilizer applied parameters. A case study based on one of the selected locations (Damphala Tea Factory) is presented here. The novelty of the presented work aligns with the following.

- It introduces the first comprehensive artificial intelligence model designed to predict tea crop yields in Sri Lanka.
- It employs explainable artificial intelligence techniques to clarify the black-box nature of conventional machine learning approaches.

2. Study area and data

2.1. Dampahala Tea factory

Due to the data limitations and restrictions within the context of Sri Lanka, Dampahala Tea Company (Pvt) Ltd was selected for the case study application. It was established in 1984, and is located at coordinates 6° 16′ 8″ N, 80° 37′ 50″ E in Heegoda, Urubokka within the Matara District of Sri Lanka's Southern Province. The factory runs from the raw tea leaves primarily collected from smallholder farmers who own plots ranging from 1 to 50 acres. The company collaborates with around 1400 smallholders and 10 medium-sized tea collectors. These collectors, who work closely with the small farmers, contribute approximately 23% of the monthly raw tea intake, while the remaining 77% comes directly from the farmers themselves. The company's monthly target for raw tea collection is 200,000 kg, largely gathered from tea-growing areas such as Bengamuwa, Wilayaya, Urubokka, Rammalakanda, Udapasgoda, and Dampahala. The location of the factory and the tea plucking areas are shown in Fig. 1.

The varieties of tea cultivated in the area include T 2024, 2025, 2023, and 2043, which are particularly suited to the monsoonal climate of Southern Sri Lanka, yielding a unique flavour profile recognized worldwide as Ceylon black tea (orthodox black teas). This distinctive flavour has earned Southern Sri Lanka tea a celebrated place on the global stage. The transformation from raw to made tea follows a precise process, resulting in a final output of approximately 40,000 kg of finished product each month. These finished products include a range of popular tea grades such as OP, OP1, OPA, BOP, PACO, PACO1, and other specialty teas specific to Sri Lanka. Dampahala Tea Company utilizes around 150 tons of fertilizer annually to maintain high-yield quality, with about 60% of this supply classified under the U750 fertilizer category. This fertilization approach, combined with the region's climate, supports the tea's robust growth and enhances the distinctive taste and aroma associated with Southern Sri Lanka's black tea.

2.2. Data

The data required to develop the prediction model were collected from Dampahala tea factory and the Department of Meteorology, Sri Lanka. Meteorological data including monthly rainfall (mm), evaporation (mm), sunshine hours, relative humidity (mm) at 8.30am and 3.30pm, soil temperatures at 10 cm and 30 cm depths at 8.30am and 3.30pm and applied fertilizer data including T0 200 (kg), T0 750 (kg), T 0 1240 (kg), U 0 709 (kg), U 0 834 (kg), Yara Mila (kg), Dolomite ($CaMg(CO_3)_2$) (kg), BioC fertilizer (kg), compost (kg), UT0752 (kg), UT01625 (kg) were obtained monthly from January 2013 to August 2023. In addition, the total fresh green tea crop rate (kg/ha) was also collected for the same period. Table 1 presents the selected data as the inputs for the developed model.

These selected factors (RF, E, SH, RHM, RHE, STM, STE, F1–F11) are critical for analyzing tea yield as they capture key environmental and nutrient management influences on tea plant growth. Rainfall, evaporation, sunshine, humidity, and soil temperature directly affect water availability, photosynthesis, and root health of the tea plant. Moreover, fertilizers (F1–F11) optimize nutrient supply for leaf production. These factors were likely chosen based on prior research, local expertise, and experimental design to model their combined impact on yield. This comprehensive selection enables the precise identification of conditions and inputs that maximize tea productivity.

3. Methodology

A detailed methodology for predicting fresh green tea crop rate with respect to the meteorological and nutrition applied to soil layers is given in this section. The following steps were carried out to mathematically model the prediction Eq. (1).

Fig. 1. Dampahala tea factory and sourcing area.

Table 1
Selected data for model development.

Meteorological data	Fertilizer data
Monthly rainfall (*RF*)	T0 200 (F_1)
Evaporation (*E*)	T0 750 (F_2)
Sunshine hours (*SH*)	T 0 1240 (F_3)
Relative humidity at 8:30AM (RH_M)	U 0 709 (F_4)
Relative humidity at 3:30PM (RH_E)	U 0 834 (F_5)
Soil temperature at 8:30AM (ST_M)	Yara Mila (F_6)
Soil temperature at 3:30PM (ST_E)	Dolomite (F_7)
	BioC fertilizer (F_8)
	Compost (F_9)
	UT0752 (F_{10})
	UT01625 (F_{11})

$$Fresh\ Tea\ Crop\ Rate = Function(RF, E, SH, RH_M, RH_E, ST_M, ST_E, F_1, F_2, F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}) \tag{1}$$

3.1. Data preprocessing

The collected data had some variations which could impact negatively on the predictions. These variations may have resulted from sensor errors, machinery malfunctions, or human mistakes during the data collection and recording processes. An unlearned dataset has the capability of developing some unrealistic relationships [42]. Therefore, data preprocessing is a requirement. Data preprocessing is an important process for most of the numerical data-based analysis [43]. Data points with significant deviations were removed from the dataset, and the patterns associated with the varying parameters were analyzed using appropriate plots.

3.2. Correlation matrix development

Pearson correlation is used to understand the linear relationship between two datasets [44]. This is helpful in feature selection for the training of the models. Feature selection is one of the crucial steps in prediction models using machine learning concepts [45]. The highest impactful input parameters were identified for the model development. In addition, the linearity and non-linearity can be identified from the correlation matrix.

3.3. Noise addition to the data

Noise is an undefined signal wave that is connected with the original signal. It can positively or negatively impact the time series [46]. The datasets used in this research were augmented using Gaussian noise. Overfitting the training data is one of the main issues in cutting-edge AI models. This is mainly due to the small fluctuations and the patterns of the dataset. However, the addition of a well-known calibrated noise would help not to be sensitive to outliers or other data fluctuations that are not in the original data patterns [46].

Regularization effect, smoothness of the function, increase of the data variability, and reduce the sensitivity to the noisy fluctuations in the real-world data are some main key advantages of using Gaussian noise over the real-world data. Eq. (2) showcases the development of the Gaussian function.

$$N \sim N(\mu, \sigma^2) \tag{2}$$

where μ is the average value and σ is the standard deviation of the noise. Eq. (3) presents the mathematical formulation developed after the addition of noise [47].

$$X_{Noisy} = X_{Original} + \alpha.N \tag{3}$$

where X_{Noisy} is the final output data with the added noise, $X_{Original}$ is the original real-world data, and α is a predefined scalling factor that is used as the scaling factor. This scaling factor controls the noise level.

3.4. Correlation after the noise addition

After the noise addition, the improvement of the dataset was examined using the correlation matrix again. The main contributing factors were re-examined. This illustrates whether the slight changes that happened to the dataset are acceptable or not. Following the addition of Gaussian noise, we found that the positive correlation among the contributing factors had increased slightly.

3.5. Training of the models using cutting edge machine learning models

Seven cutting-edge machine learning models were utilized to carry out predictions using the formulation given in Eq. (1). The details of these machine learning models are given in the following sections. The machine learning algorithms used in this study include XGBoost, CatBoost, LightGBM along with ANN, Ridge, LASSO and ElasticNet. These models exhibit effective capabilities for complex relation learning and multiple data type management and strong prediction outputs. These algorithms demonstrate high predictive power which helps identify advanced patterns between environmental parameters (such as rainfall) and fertilizer elements (F1–F11) since these components affect tea yield variations significantly. The algorithms possess strong resistance against overfitting issues through LASSO or ElasticNet regularization approaches and XGBoost and LightGBM tree-based algorithms that guarantee reliable results from potentially noisy data sets.

1. Extreme Gradient Boost Algorithm (XGBoost)
   Extreme Gradient Boost (XGBoost) Algorithm has performed well on the regression applications. It has the capability of minimizing its own errors. Each new iteration is trained by eliminating the error that happened in the previous decision tree. Therefore, the accuracy of the training process increases. In addition, the outliers are effectively handled in XGBoost [44].
2. Categorical Boost Algorithm (CatBoost)
   This algorithm follows a gradient boosting framework which is optimized for the categorical data efficiently. More into the functionality, the CatBoost algorithm follows an ordered boosting method which has the capability of handling the target leakage and calculating the statistics for the categorical features. This enhances the generalization capability of the model. Furthermore, this can handle the model overfitting more effectively. Therefore, the computational time of the model is very low compared to the other boosting algorithms [48].
3. Light Gradient Boosting Algorithm (LightGBM)
   LightGBM is an open-source gradient-boosting decision tree algorithm that was designed to perform well on large-scale data. Based on the histogram algorithm, it uses a parallel voting decision tree to accelerate the training speed with lower memory use, supporting complex network connectivity to improve the capability for parallel learning [44]. Unlike level-wise growth, LightGBM splits the leaf with the maximum gain. Hence, it is more focused on sections of data that would provide the largest improvement in the accuracy of the system. This leads LightGBM to capture complex patterns in fewer iterations compared to conventional boosting algorithms. Besides optimized handling of big datasets and complex data structures, superior results are achieved without overfitting.
4. Artificial Neural Network (ANN)
   ANN is a computation model that is capable of functioning well in comparison to regression models. Similar to Long short-term memory (LSTM), ANN follows the same architecture of the input, hidden, and output layer structure. Calculation of the weights and hyperparameter tuning are crucial steps that have to be followed in the ANN. One of the major drawbacks of the ANN is the overfitting of data [49]. Therefore, proper tuning of the hyperparameters is essential. Thus, grid search and cross-validation were applied over the ANN model to verify the proper functionality of the model in this research.
5. Ridge Regressor
   Ridge Regression is an advanced technique developed to address overfitting and multicollinearity in linear regression models. It modifies the traditional Ordinary Least Squares (OLS) regression by introducing a regularization term (L_2), which penalizes large coefficients in the model. This regularization term is proportional to the squared L_2 norm of the coefficients, effectively shrinking them towards zero but not exactly to zero, ensuring stability in the model [50]. The L_2 regularization term helps to mitigate issues of multicollinearity, where independent variables are highly correlated. In such scenarios, OLS regression can lead to coefficient inflation, causing instability and large variations in the coefficients. Ridge regression controls this by adding a penalty, reducing the impact of highly correlated variables and resulting in more reliable, stable coefficient estimates. This approach is particularly beneficial when dealing with high-dimensional data or when the number of features exceeds the number of observations.
6. Least Absolute Shrinkage and Selection Operator (LASSO)
   Lasso Regression is one of the most powerful methods in machine learning approaches; it is mainly used for regression tasks working with a high-dimensional dataset [50]. Using L_1 regularization, it penalizes the absolute values of regression coefficients, hence the resulting model is sparse with some of the model's coefficients set to zero. This becomes particularly useful when the dataset contains a lot of irrelevant or redundant features since LASSO performs well in selecting the most relevant variables for prediction. On the other hand, Ridge Regression uses L_2 regularization, which is a shrinkage in the coefficients but not a removal, thus being more suitable for multicollinearity contexts where it is assumed that all features contribute to the outcome. LASSO is ideal for feature selection tasks because it can easily generate a simpler and more interpretable model with fewer variables. In contrast, Ridge is applicable in cases when there are highly correlated predictors, and all features need to be kept. Both techniques prevent overfitting and control model complexity, though LASSO also has sparsity, which leads to even more concise and generalizable models.
7. ElasticNet
   ElasticNet is a powerful regularization technique that overcomes the limitations of both LASSO and Ridge regression by taking the best from each. It uses L_1 regularization from LASSO for feature selection

and L_2 regularization from Ridge to handle multicollinearity. Thus, it is a very general approach when the dataset has features with sparse and correlated characteristics. In contrast to LASSO, which arbitrarily picks one feature from groups of correlated variables, ElasticNet allows for a more balanced handling of the correlated features in a manner that keeps important variables and does not drop them prematurely. Moreover, the combination of both regularization techniques, ElasticNet proves to handle high-dimensional data better and therefore is more robust, which in general improves the generalizability of the model [51].

3.6. Optimization of machine learning models

Hyperparameter tuning was carried out over the models to identify the best model. The encountered error was minimized through the optimization of the machine learning models. Moreover, the common issues such as overfitting and underfitting of the models are minimized. Overall, optimization is essential to refining the model's accuracy, efficiency, and reliability, ensuring it performs well on unseen data. Following a GridSearch mechanism the state-of-the-art models were optimized through the evaluation. Using the evaluation matrices, regression plots and training/testing curves supported the hyperparameter tuning of the models to attain the maximum capable results. The modeling was utilized and optimized on a computer, an M2 Neural Engine processor (8 cores) with 16GB RAM. The optimization architecture used is given in Fig. 2.

3.7. Testing, validation and performance evaluation of the developed machine learning models

The machine learning models were trained, tested and validated for 70%, 20% and 10% respectively. The performance evaluation of the developed models is one of the crucial steps. The comparative analysis was carried out based on the evaluation matrices including coefficient of determination (R^2), root mean squared error (RMSE), mean squared error (MSE), and mean absolute error (MAE). The best performing model was identified by the models that attained the highest coefficient of determination and the lowest error.

3.8. Explainability of machine learning models

The explainability of machine learning models is an important aspect of understanding what is happening from input to output [52]. The proposed prediction model is expected to help the decision-making in tea yield not only in the case study area but also expand to the whole country. Therefore, the best performing model was evaluated for its explainability. Shapley Additive exPlanations (SHAP) is a popular method to explain the output of a model based on the contribution of each feature in Eq. (1).

3.9. Overall methodology

The overall methodology of the prediction of tea yield based on the mathematical formulation given in Eq. (1) is shown in Fig. 3. This is the first-ever study which was carried out in the context of Sri Lanka for the world's best tea.

4. Results and discussion

As it was stated in the methodology section, several cutting-edge machine learning models were developed to predict the tea yield. The Pearson correlation matrix for the dataset used in this research work is showcased in Fig. 4. As per Fig. 4, it is noted that the linear relationship among features of the parameters is not significant. Nevertheless, the matrix does not showcase any non-linear relationships among the features.

Fig. 5 presents the coefficient of determination (R^2) for the predicted tea crop yield to the actual tea crop yield from the tested machine learning algorithms. Among them, XGBoost, CatBoost, and LightGBM demonstrated superior prediction performance, with CatBoost achieving the highest accuracy, boasting an R^2 value of 0.9. In contrast, the other algorithms, which include Artificial Neural Networks (ANN), Ridge Regression, LASSO, and ElasticNet, did not perform as well with the dataset used in this study.

Table 2 presents the performance of the tested models with respect to the other assessed indices. The results indicate a mix of performances

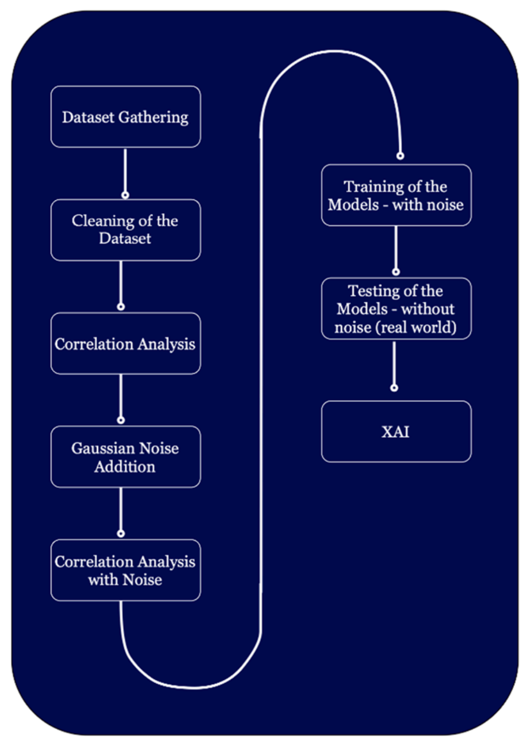


Fig. 3. Flowchart of the prediction model development.

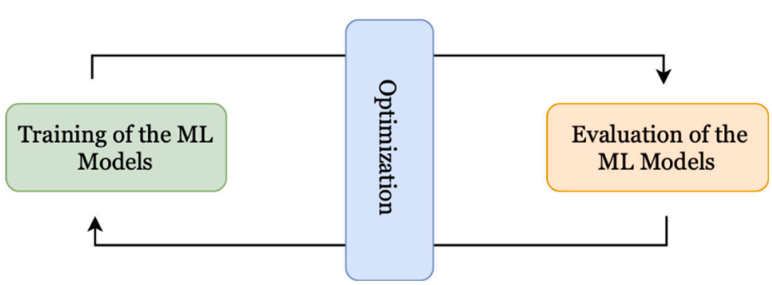


Fig. 2. Optimizing approach of the ML models.

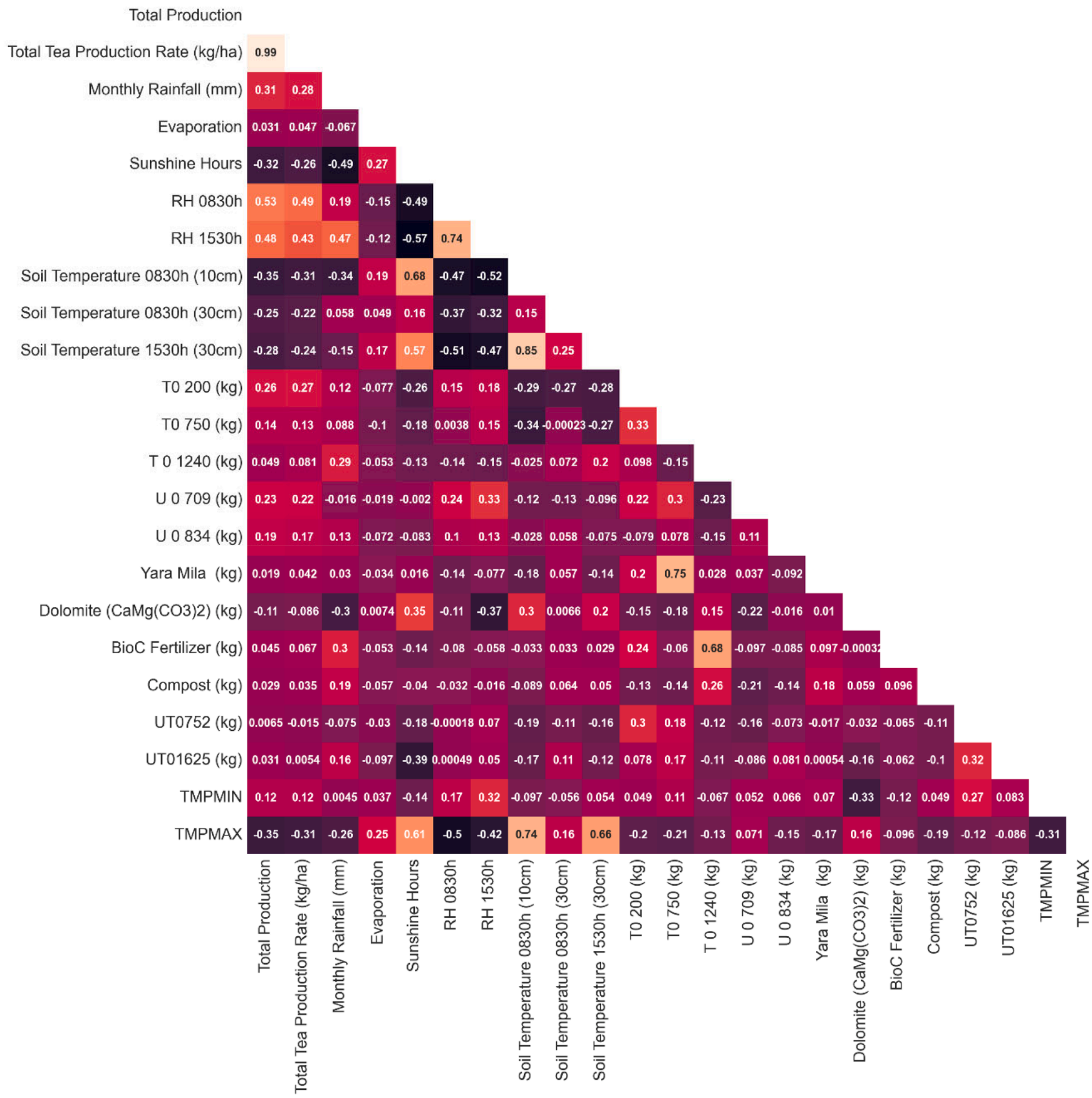


Fig. 4. Pearson Correlation Coefficient matrix.

among the models. CatBoost stands out, achieving the lowest errors in RMSE, MSE, and MAE compared to the others. Even though XGBoost and LightGBM have higher coefficients of determination, they have showcased slightly higher errors. Therefore, an explainability analysis was carried out on the CatBoost model results.

For the transparency and usability of the model over the application, SHAP waterfall diagrams, summary plots, and dependance plots were analyzed. Fig. 6 presents these SHAP analysis outcomes.

Figs. 6a and 6b clearly showcase that the Relative humidity at 8.30am, Evaporation, T0 200 fertilizer, and soil temperature at 3.30pm at 30 cm depth had a significant positive impact on the model prediction. This suggests that these features were associated with higher tea yields by making interrelationships between them. The maximum atmospheric temperature, U0 834 fertilizer, and soil temperature at 8.30am at 10 cm level had the largest negative impact on the model's predictions. This suggests that these features were associated with lower tea yields. However, it is well noted that these environmental features and soil temperatures are impossible to control to increase the tea crop yield. Nevertheless, these parameter identification suggests the importance of meteorological parameters in crop yield. In other words, the impact of climate change is significant to the tea crop yield.

Fig. 6c presents the Sample dependence plot of relative humidity at 8.30am. It showcases a potential positive relationship between RH at 8.30am to model prediction. When the RH increases there is a trend of increasing SHAP values indicating the morning relative humidity may contribute to higher yield. In addition, this dependence was plotted with the interaction to Dolomite fertilizer application. Even at a lower Dolomite application (around 1000 kg/ha), a higher yield can be observed.

Fig. 6d presents the waterfall plot of the prediction of tea crop yield with respect to the considered features. The contribution of each parameter is clearly shown here. The evaporation and T0 200 (kg) had the largest positive impact on the model's prediction, increasing it by 12.32 and 17.74, respectively. Dolomite ($CaMg(CO_3)_2$) (kg) had a significant negative impact, decreasing the prediction by 10.13. Furthermore, the plot shows a baseline value of $E[f(X)] = 664.144$. This is the starting point of the plot representing the average model prediction. The final model for the prediction for this instance is 681.367. Therefore, the prediction has higher accuracy.

Therefore, the results clearly showcased the importance of the application of machine learning algorithms in predicting tea crop yield. CatBoost algorithm was well optimized for the case study and explainability using SHAP showcases the performance and the suitability. Similar studies in the application of machine learning to predict the tea

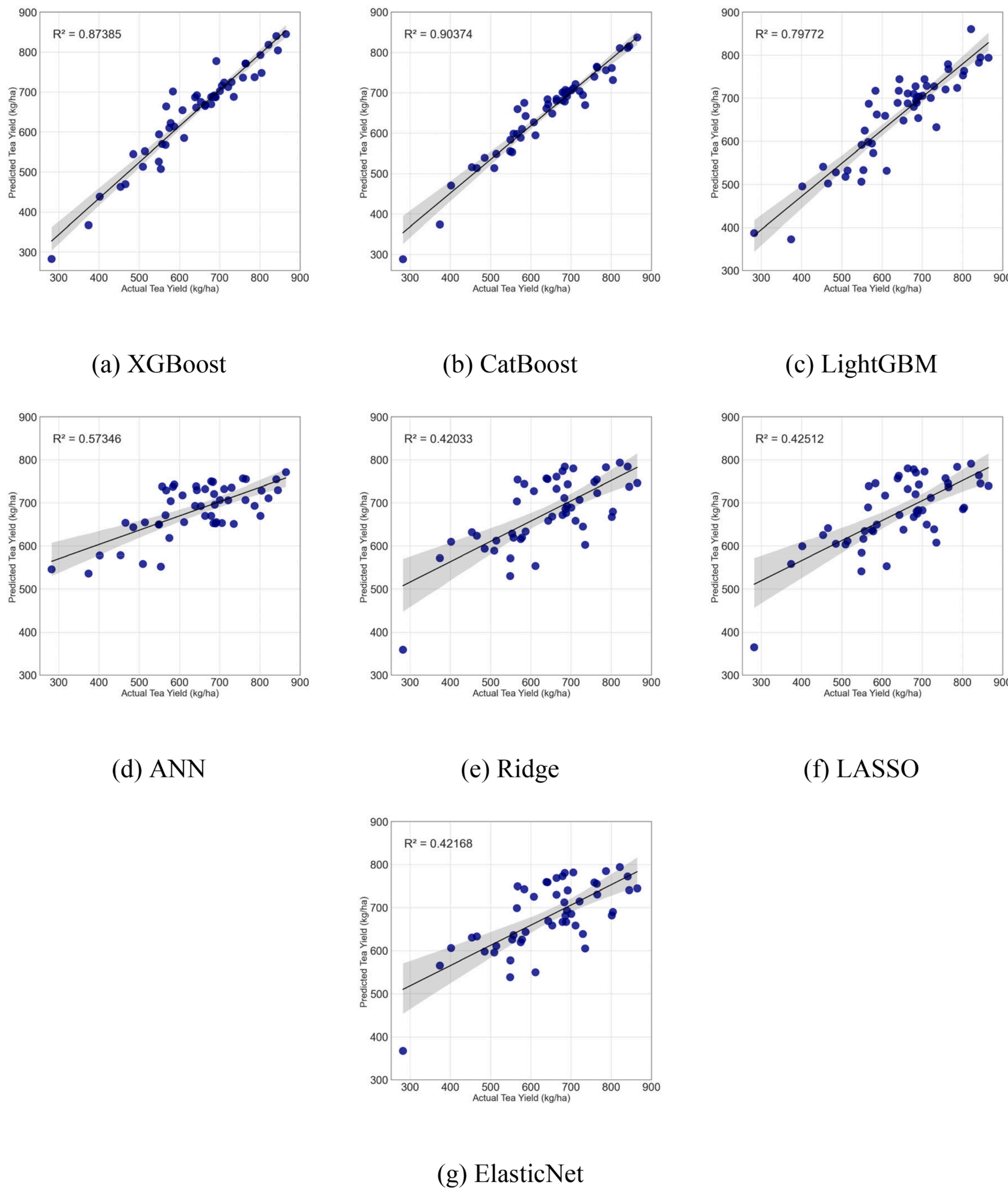


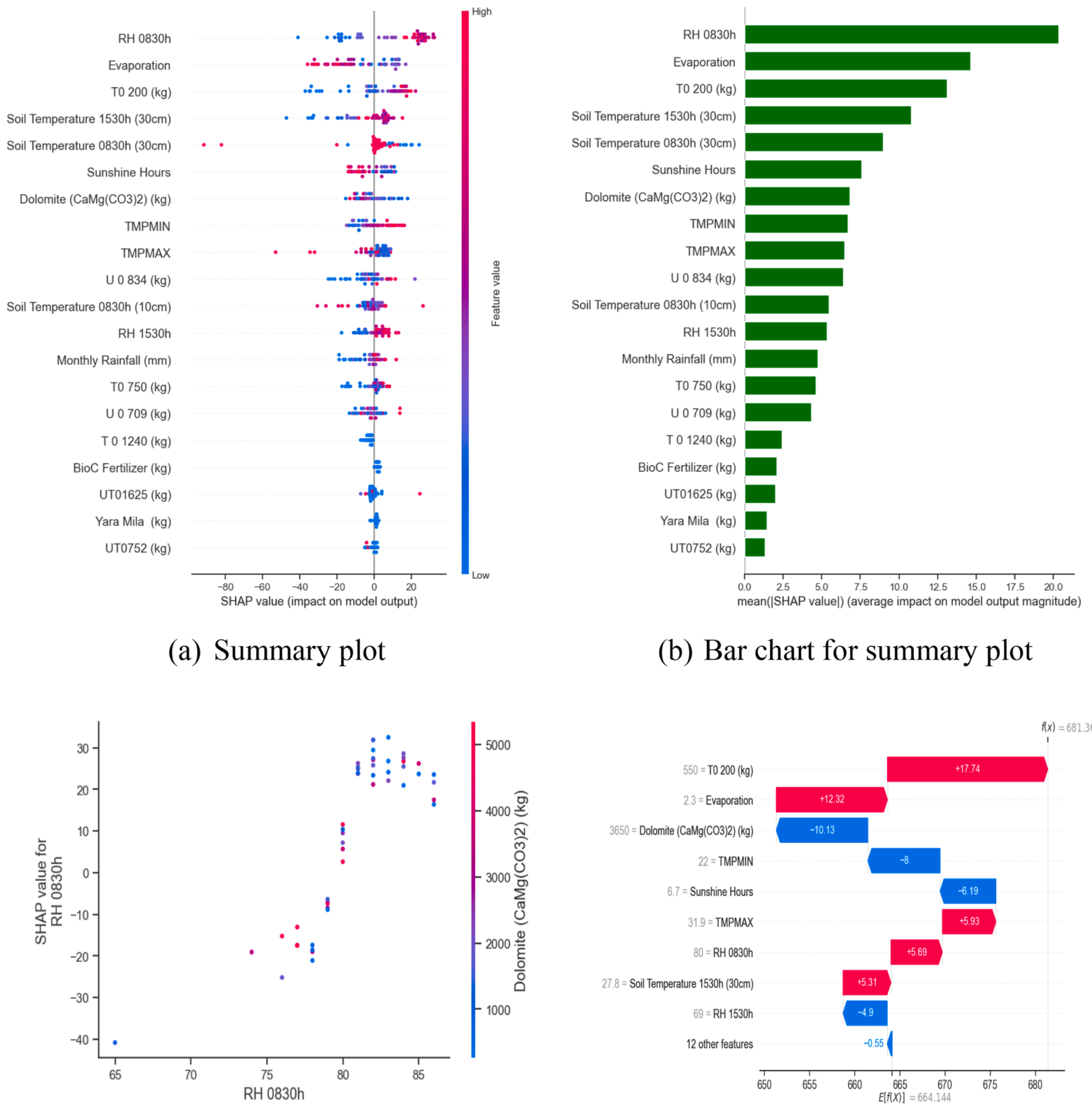
Fig. 5. Regression curves for prediction of Total Tea Yield Prediction.

Table 2
Performance of tested machine learning algorithms.

Model	R^2	RMSE	MSE	MAE
XGBoost	0.87385	44.209	1954.479	31.103
CatBoost	**0.90374**	**38.617**	**1491.324**	**28.104**
LightGBM	0.79772	55.982	3134.079	41.787
ANN	0.57346	81.293	6608.689	65.462
Ridge	0.42033	94.769	8981.296	76.114
LASSO	0.42512	94.377	8907.065	77.585
ElasticNet	0.42168	94.658	8960.158	76.666

crop yield can be found in the literature. Batool et al. [24] have showcased that the XGBoost regressor outperformed the tested others with the lowest possible error in their study. Similarly, a number of other research papers can be found on tea crop yield prediction. However, none of them were to the world's best tea producer, Sri Lankan tea. In addition, explainability aspects were never considered in the black machine learning models. Therefore, this research has overcome those identified research gaps.

However, the research presented here has the limitation of a case study. Therefore, the methodology has to be expanded to the whole country. With the availability of data from all tea estates in Sri Lanka



(a) Summary plot

(b) Bar chart for summary plot

(c) Sample dependence plot for RH 0830h

(d) Waterfall plot

Fig. 6. SHAP analysis.

with the collaboration of meteorological data, a comprehensive but universal model can be presented. The model can be equipped with the type of tea, grown area, etc.

As it was stated in the introduction, tea production is one of the major income sources in Sri Lanka. Therefore, prediction of the tea crop production to a higher accuracy is highly important. This research showcases the first-ever such predictive model to Sri Lanka by a case study application. The model accuracy developed here showcased the applicability of the model on a global scale. Therefore, the prediction model can be trained across the country to predict the tea crop yield in Sri Lanka. Therefore, the model would help the authorities to move forward with policy decisions to achieve sustainable development goals. In addition, the model can be used in future climatic scenarios (like shared socioeconomic pathways) to forecast the yield of tea crops in the future. According to the predictions, Sri Lanka is going to be impacted significantly by climate change. It is a well-known fact that the change in climate cannot be reversed. However, there is a possibility to introduce climate resilience tea crops in Sri Lanka. This research would be significantly helpful in order to understand the climatic parameters which are influencing the tea crop yield. Therefore, further research can be carried out in the direction of introducing climate resilience tea crops with the help of the findings of this research.

5. Conclusions

The first-ever tea crop prediction model for Sri Lankan Tea has showcased its applicability by a case study application. CatBoost machine learning algorithm has outperformed all other tested models with higher accuracy. Therefore, a solid prediction model is presented herein with the environment of climate change. In addition, this is the first-ever tea prediction model solidified with the explainability aspects of machine learning algorithms using SHAP analysis. Feature identification has showcased that the meteorological parameters of morning relative humidity and evaporation have made a significant contribution to the prediction. In addition, fertilizer T0 200 has also showcased its significance.

Sri Lanka as one of the major tea-growing countries in the world, should have to understand its role in changing climate. It is one of the most climate-impacted countries in the world. Therefore, accurate predictions for foreign income-earned crops are an essential task to the future world. This reinforces the expansion of the developed model to a countrywide model development.

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CRediT authorship contribution statement

Lakindu Mampitiya: Writing – original draft, Software, Methodology, Investigation, Formal analysis. **Harindu S. Sumanasekara:** Writing – original draft, Data curation. **Namal Rathnayake:** Writing – review & editing, Validation. **Yukinobu Hoshino:** Writing – review & editing, Validation, Formal analysis. **Upaka Rathnayake:** Writing – review & editing, Validation, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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